

The Matrix Multiplication Bound

$\omega \leq 9/4$ over Associative Rings

Abstract

OpenAI’s auxiliary-separation argument gives the matrix multiplication exponent bound $9/4$ over the complex numbers [1, 2]; our earlier extension establishes the same bound over every field [3]. We describe an integral version of that argument. Its stronger intermediate conclusion is a bound on exact tensor rank over $\mathbb{Z}$, which supplies the same exponent upper bound over every associative unital ring. The zero ring is included using an explicit zero-cost program and the real-infimum convention in the formal definition. The changes are a primitive tensor semiring for spectral detection, coefficient extraction and finite free descent, and coprime patching of unnormalized Fourier and interpolation constructions. The numerical exponent and the underlying profile-growth argument are those of OpenAI.

Verification status. The full upper-bound theorem, including the zero ring, has compiled and passed its axiom audit, fresh-kernel replay, and frozen-specification Comparator with all bundled checkers. These checks validate formal declarations, not this prose document.

AI disclosure. This manuscript is AI-generated. The proof development and Lean formalization were produced with Codex under human direction. Attribution to the underlying OpenAI proof and the previous field extension is given below; no claim of independent numerical improvement or human peer review is made.

1 Statement and contribution

For an associative unital ring R , define $\omega(R)$ using division-free arithmetic programs: addition, subtraction, and multiplication each cost one operation; input and constant gates cost zero. The order of factors in a multiplication gate is significant. An exponent τ is admissible when

$$\forall \varepsilon > 0 \exists C > 0 \forall n \geq 1 \exists P_n : \quad P_n \text{ computes matrix multiplication over } R, \quad \text{cost}(P_n) \leq Cn^{\tau+\varepsilon}.$$

Correctness means correctness for every pair of input matrices of the specified size. The exponent is the real-valued $\mathbf{sInf}$ of admissible exponents, with the convention for the zero ring explained below. This is the arithmetic model used in the OpenAI Lean development, with its scalar type generalized from a field to a ring.

Write M_n for the square matrix multiplication tensor and $R_{\mathbb{Z}}(T)$ for exact tensor rank with integer coefficient vectors. Put

$$\nu_{\mathbb{Z}} = \inf_{n \geq 2} \log_n R_{\mathbb{Z}}(M_n).$$

A tensor identity here is equality at every coefficient, not merely agreement of the corresponding functions over finite fields.

Theorem 1.1. *The exact integer coefficient rank exponent satisfies $\nu_{\mathbb{Z}} \leq 9/4$. Consequently, for every associative unital ring R and every $\varepsilon > 0$, there are division-free arithmetic programs computing square matrix multiplication with cost $O_{R,\varepsilon}(n^{9/4+\varepsilon})$. For every such R , the formally defined exponent satisfies*

$$\omega(R) \leq \frac{9}{4}.$$

Remark 1.2 (The zero ring and the real infimum). In the zero ring, one free constant-zero register supplies every output, giving a correct program of cost zero. Every real exponent is therefore admissible. The admissible set is $\mathbb{R}$, which is not bounded below; Mathlib’s real-valued `sInf` convention gives $\omega(R) = 0$. The extended-real infimum would instead be $-\infty$. For nontrivial rings, admissible exponents are bounded below by 2, so the usual infimum interpretation applies. The integral argument handles that case; the zero-ring case uses the explicit program and the stated convention.

OpenAI supplies the complex-field bound, auxiliary separation, determinant and sector constructions, and the real profile-growth argument [1, 2]. The previous field extension changes the Fourier period in positive characteristic and descends from an algebraic closure through a fixed finite extension [3]. We do not infer Theorem 1.1 from the existence of solutions over all fields. Instead we construct and patch integer coefficient identities. This also handles rings with zero divisors, nilpotents, and noncommutative multiplication. The coefficient algebras used in the proof remain commutative.

2 The integral spectral setting

Write $U \preceq T$ if three integer linear maps restrict T to U . Pass to tensors modulo mutual restriction, with direct sum as addition and tensor product as multiplication. Let $\mathcal{S}$ be the subsemiring consisting of zero and the classes satisfying $1 \preceq T$. We call these classes *unit-dominating*. A tensor with a coefficient 1 or -1 is unit-dominating.

This restriction on the spectral setting matters. The nonzero scalar tensor with coefficient 2 does not restrict over $\mathbb{Z}$ to the scalar tensor 1: such a restriction would give $2abc = 1$ for integers a, b, c . Thus nonzero tensors cannot simply be treated as positive spectral objects in the field proof.

For nonzero $T \in \mathcal{S}$ one has

$$1 \preceq T \preceq \langle R_{\mathbb{Z}}(T) \rangle,$$

where $\langle m \rangle$ denotes the diagonal tensor of m copies of 1. These comparisons give precisely the cofinality and positivity used by the detecting-character argument of [1, Appendix A]. The order of diagonal tensors is the ordinary order of their sizes, as follows from flattening over $\mathbb{Q}$. Flattening gives $\nu_{\mathbb{Z}} \geq 2$; the naive multiplication scheme gives $\nu_{\mathbb{Z}} \leq 3$.

The detecting-character proof therefore runs on $\mathcal{S}$: for $d \geq 2$ and an integer $0 \leq k < d^{\nu_{\mathbb{Z}}}$, there is a normalized, additive, multiplicative, restriction-monotone character φ with $\varphi(M_d) \geq k$. Its compactness, rational separation, and fixed-point steps are inherited from [2]; every division by a character value is justified by $\varphi(T) \geq 1$ for nonzero $T \in \mathcal{S}$. The catalytic rank argument extracts 1 only from such an object.

For convenience, these characters extend to all tensor classes:

$$\chi(T) = \varphi(T \oplus 1) - 1. \tag{2.1}$$

Since $T \oplus 1$ is unit-dominating, this is defined everywhere. Additivity and multiplicativity follow by adjoining copies of 1 and expanding $(T \oplus 1) \otimes (U \oplus 1)$; monotonicity and nonnegativity follow from

restriction. The extension agrees with φ on $\mathcal{S}$. In particular it preserves the detecting value on M_d . This avoids asserting positivity for every nonzero integer tensor.

3 Transport of exact coefficient identities

The following two elementary constructions replace field-valued interpolation in the treatment of polynomial degenerations.

Lemma 3.1 (Fixed finite free descent). *Let E be a commutative $\mathbb{Z}$ -algebra free of finite rank D , with a $\mathbb{Z}$ -linear map $\pi : E \rightarrow \mathbb{Z}$ satisfying $\pi(1) = 1$. If integer tensors U, T satisfy $U \preceq_E T$, then*

$$U \preceq_{\mathbb{Z}} \bigoplus_{D^2} T.$$

The same overhead D^2 applies after taking any tensor power in this one fixed algebra E .

Proof. Choose a basis $e_1, \dots, e_D$ of E . Expand the first two local maps as $A = \sum_i A_i e_i$ and $B = \sum_j B_j e_j$, with integer coefficient maps A_i, B_j . For each pair (i, j) apply $c \mapsto \pi(e_i e_j c)$ to the third local map. Summing the resulting D^2 integer restrictions recovers U , because π splits integer coefficients. Taking powers before this expansion retains D^2 ; descending first and then taking powers would instead incur D^{2k} . $\square$

Lemma 3.2 (Coefficient extraction). *Suppose polynomial local maps $A(t), B(t), C(t)$ of degrees at most a, b, c transform T into a polynomial tensor whose coefficient of t^h is U . Then U is a restriction of $(a+1)(b+1)$ copies of T . For a degeneration with leading coefficient U , its k th tensor power has extraction cost at most $(ka+1)(kb+1)$.*

Proof. The required coefficient is the sum of the restrictions with maps A_i, B_j, C_{h-i-j} for $0 \leq i \leq a$ and $0 \leq j \leq b$. A negative coefficient index contributes zero. Powers multiply the leading coefficient and scale each degree bound by k . $\square$

Both lemmas hold over commutative base rings with the corresponding finite free data. No evaluations or division are needed. Monic quotients $E = \mathbb{Z}[t]/(f)$ supply the needed map π : reduce to degree below $\deg f$ and take the constant coefficient, assuming $\deg f > 0$.

We also use a literal Bezout patch. If $d_1, \dots, d_s$ generate the unit ideal in $\mathbb{Z}$, their k th powers do too. Choose integers u_j with $\sum_j u_j d_j^k = 1$. Concatenating rank decompositions for $d_j^k T^{\otimes k}$, and multiplying one local map by u_j , gives a decomposition for $T^{\otimes k}$. This is a coefficient identity; it does not divide by any d_j or treat scalar multiplication as a number of direct-sum copies.

4 Integral convolution and Fourier separation

4.1 Interpolation with coprime norms

For positive integers a, b , let $C(a, b)$ be the polynomial multiplication tensor with coefficients $[i+j=\ell]$, and put $h = a+b-1$. The required statement is an asymptotic rank bound, not a rank- h decomposition over $\mathbb{Z}$ at the first power.

Proposition 4.1. *For each a, b there is a positive integer $K_{a,b}$ such that*

$$\mathrm{R}_{\mathbb{Z}}(C(a, b)^{\otimes k}) \leq K_{a,b} h^k \quad (k \geq 0).$$

Every normalized nonnegative character monotone under restriction obeys $\chi(C(a, b)) \leq h$.

Proof. Fix a prime p . Choose a finite extension of $\mathbb{F}_p$ with at least h elements and a power basis, then lift its monic minimal polynomial to a monic $f \in \mathbb{Z}[t]$. The quotient $E = \mathbb{Z}[t]/(f)$ is finite free of some positive rank D , and its basis specializes to the finite-field basis. Choose lifts $v_1, \dots, v_h$ of distinct residue elements. Write V for their Vandermonde matrix and $\Delta = \det V$.

Evaluation followed by $\text{adj}(V)$ gives an h -term scheme for $\Delta C(a, b)$ over E . Let M_Δ be the integer matrix of multiplication by Δ in the chosen basis, and put $d = \det M_\Delta$. If w is the element with coordinate vector $\text{adj}(M_\Delta)[1]$, then $\Delta w = d1$. Multiplying the output map by w therefore gives an h -term scheme for $dC(a, b)$ over E , without passing to a localization.

Modulo p , the nodes remain distinct. Thus Δ is nonzero in the residue field, multiplication by it is invertible, and $p \nmid d$. Taking powers in this fixed E and applying Lemma 3.1 gives

$$R_{\mathbb{Z}}(d^k C(a, b)^{\otimes k}) \leq D^2 h^k.$$

Perform this construction for each prime. No prime divides all resulting clearing integers, so the ideal they generate is $\mathbb{Z}$. Membership of 1 in that ideal uses only a finite subfamily $d_1, \dots, d_s$. Fix this subfamily and its algebras before varying k . The Bezout patch gives the assertion with $K_{a,b} = \sum_j D_j^2$. Taking k th roots of the character inequality removes this fixed constant. $\square$

4.2 Two unnormalized Fourier periods

For $M \geq 1$ matched sectors, take $L = 5M$ and use each period $q \in \{L, L+1\}$. Work in the finite free cyclotomic algebra $E_q = \mathbb{Z}[z]/(\Phi_q(z))$, with ζ the image of z . It is a unit and satisfies the unnormalized identity

$$\sum_{r=0}^{q-1} \zeta^{re} = \begin{cases} q, & q \mid e, \\ 0, & q \nmid e. \end{cases}$$

Apply the substitutions of [1, Proposition 3.1], omitting their factor $1/q$. The relevant phases have absolute value at most $3(M-1) < q$, so the same matched terms survive. The relative monomial weights select the same leading tensor, multiplied by q . To make all local exponents nonnegative, add M^2 to each second- and third-leg weight. The selected coefficient moves to degree $2M^2$; its tensor is unchanged.

Let A be the shared-first-leg source and B its separated target, including the auxiliary dot-product factor. The construction gives a polynomial degeneration from q copies of A to the coefficient multiple qB over E_q . Taking powers first, then extracting and descending, gives

$$q^k B^{\otimes k} \preceq_{\mathbb{Z}} \bigoplus_{D_q^2 P_q(k) q^k} A^{\otimes k},$$

where $D_q = \deg \Phi_q$ and P_q is a fixed quadratic polynomial bound. The q^k on the left multiplies coefficients; on the right it counts copies. Since L^k and $(L+1)^k$ are coprime, patch the two restrictions before taking character values. Then

$$\chi(B)^k \leq (D_L^2 P_L(k) L^k + D_{L+1}^2 P_{L+1}(k) (L+1)^k) \chi(A)^k.$$

Taking roots yields $\chi(B) \leq (L+1) \chi(A) \leq 6M \chi(A)$. The constant six replaces the original constant five and disappears in the entropy limit. For each fixed outer type length, the sector count M , both algebras, and all degree bounds are fixed before the inner power k tends to infinity. No estimate uniform in growing M is needed.

5 The retained profile argument and ring algorithms

The determinant filtration uses an integral adapted basis. On its affine chart, for pairs $(A(X), B(X))$ of degree at most e , take quotient vectors $(X^i, 0)$ for $0 \leq i \leq e$ and $(0, X^e)$, and kernel vectors $(-X^{j+1}, X^j)$ for $0 \leq j < e$. The inverse reads quotient coordinates from $A + XB$ and kernel coordinates from $B_0, \dots, B_{e-1}$. These are mutually inverse integer maps; no field dimension argument is needed. The sector construction similarly uses coordinate maps and integer monomial weights.

Thus the determinant, shifted-tripling, and entropy inequalities of [1] retain their form, while Proposition 4.1 supplies the needed convolution upper bound. For the symmetrized profile P and its parameter $t > 0$, the retained laws include symmetry, row concavity, shifted tripling

$$P(a, 3b + a - 1) \geq 3P(a, b), \quad P(a, b) \leq (a + b - 1)^{1/t}.$$

OpenAI’s real diagonal-growth argument forces $t \leq 3/4$, hence $\chi(M_n) \leq n^{9/4}$. Detecting characters from Section 2 then give $\nu_{\mathbb{Z}} \leq 9/4$. This step invokes their proof rather than supplying a new profile inequality or a better numerical exponent.

Fix a block size $u \geq 2$ and an integer rank- r scheme with $r < u^{9/4+\delta}$, where $0 < \delta < \varepsilon$. Every output of the scheme is

$$\sum_{\ell=1}^r w_{c\ell} \left(\sum_a u_{\ell a} x_a \right) \left(\sum_b v_{\ell b} y_b \right), \quad u_{\ell a}, v_{\ell b}, w_{c\ell} \in \mathbb{Z}.$$

Integer constants are central in every associative unital ring. Expansion always keeps x_a before y_b , so the coefficient identity works without commuting input variables. It also works for matrix blocks, enabling the original recursive program construction. Its recurrence is $S(uv) \leq rS(v) + 6ru^2v^2$ with u, r fixed. Using $r \geq u^2$, padding, and positive exponent slack gives the operation bound in Theorem 1.1. The direct operation-count statement covers every ring. Nontriviality remains a hypothesis of the bounded-below and lower-bound statements; the zero-ring exponent is handled as in Remark 1.2.

6 Formal scope and provenance

The exported theorem in namespace `OAI.MatrixMultiplication` is

```
theorem omega_le_nine_quarters_all_rings
  (R : Type u) [Ring R] :
  Arithmetic.omega R <= (9 : Real) / 4
```

Lean’s `Ring` class here means associative unital ring, not necessarily commutative or nontrivial. The bounded-below admissibility and $\omega(R) \geq 2$ theorems retain `[Nontrivial R]`; neither holds for the zero ring. Auxiliary tensor rank and coefficient-algebra constructions are over commutative rings, with the main intermediate specialized to $\mathbb{Z}$.

The formal conclusion is existence of a family of arithmetic programs. It does not assert an $O(n^{9/4})$ endpoint bound, an efficient uniform circuit generator, or bit complexity for encodings of ring elements. In particular an arbitrary ring is not assumed to have a finite-tape representation. The integer schemes are stronger coefficient statements than functional correctness modulo a chosen characteristic.

The new repository [4] starts from the minimal import closure of the prior all-fields entry point at revision 08481ef22bca7dc9ffba091083b7c1e81e537220. Its import manifest records the inherited

source hashes and OpenAI upstream revision. The current toolchain is Lean 4.35.0-rc4 with Mathlib revision `f0469b25d97aef3998d4bc06f6f01da670b3d18e` [5, 6].

Within the matrix-multiplication subtree, the relevant files under `AuxiliarySeparation/` are:

- `Tensor/Primitive*.lean` and `Character/Existence.lean`: the spectral domain and detection;
- `Integral/`: coefficient extraction, finite free descent, norm clearing, coprime patching, Fourier separation, and the arithmetic bridge;
- `Determinant/` and `Sector/`: the integral versions of the inherited constructions.

The checked transport declarations include

```
convolution_integer_power_rank
exists_prime_clearing_convolution
```

and the polynomial degeneration `power/extract` methods.

The six audited claims use only `propxt`, `Classical.choice`, and `Quot.sound`. Their complete dependencies passed fresh-kernel replay: the upper bound for all rings, boundedness and nonemptiness of the admissible set, the lower bound, direct costs, and integer coefficient identities. Boundedness and the lower bound retain nontriviality hypotheses. Comparator checks these against a frozen model with exactly seven field-to-ring substitutions and no definition holes. Its Lean paranoid, lean4lean, nanoda, con-leche, con-ron, and default Lean checkers all accepted the strengthened statement. Verification uses trusted local sources, pinned dependencies, and incremental build caches. Comparator’s build sandbox is disabled. The repository records commands, outcomes, prior failure-control tests, and separate AI source reviews.

Attribution. The numerical $9/4$ bound and its main mathematical architecture are due to OpenAI [1, 2]. The prior field extension is credited in [3]. This manuscript concerns the integral spectral setting, explicit multiplier elimination, fixed-algebra transport, and the passage to arbitrary associative rings. Standard descent, adjugate, and Bezout principles are used without a historical-priority claim. The manuscript and the extension’s Lean proof development are AI-generated; neither is presented as independent human peer review.

References

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- [3] S. Navot, *Matrix multiplication exponent at most $9/4$ over arbitrary fields: a Lean formalization*, version 1.0.0, 2026. [doi:10.5281/zenodo.23219128](https://doi.org/10.5281/zenodo.23219128). The present source baseline is [revision 08481ef2](#) of that repository; see its provenance records for the archived release.
- [4] *Matrix multiplication exponent at most $9/4$ over associative rings: Lean development*, source repository. https://github.com/selanavot/omega_le_nine_quarters_all_rings. Source, paper, import provenance, and verification records are versioned together; the manuscript is associated with its repository commit.

- [5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction—CADE 28, LNCS **12699** (2021), 625–635. [doi:10.1007/978-3-030-79876-5_37](https://doi.org/10.1007/978-3-030-79876-5_37).
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